

TAHOUA, Niger

WMO: 610430

Lat: 14.876N

Lon: 5.265E

Elev: 1266

StdP: 14.04

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	58.6	60.7	8.1	8.7	79.1	13.9	11.7	78.6	28.7	73.1	26.4	74.8	9.0	80	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.4	107.8	68.0	106.2	68.1	104.6	67.8	78.3	92.4	77.5	91.0	77.0	90.1	7.7	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	139.4	82.1	74.6	135.6	82.1	73.8	131.9	82.4	42.8	92.3	42.0	90.9	41.4	90.3	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.5	19.5	17.4	53.9	111.2	1.9	1.3	52.5	112.1	51.3	112.9	50.3	113.7	48.8	114.6		
			40.0	83.1	1.1	2.7	39.2	85.0	38.6	86.6	38.0	88.1	37.2	90.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.3	74.7	80.0	87.0	93.3	94.2	90.7	86.3	83.8	86.4	88.3	83.0	76.0	
	DBStd	7.09	4.70	5.80	4.88	3.03	3.16	3.75	3.48	3.22	3.17	2.57	3.59	3.99	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	12895	766	841	1147	1300	1371	1221	1125	1048	1093	1186	991	806	
	CDD65	7420	301	421	682	850	906	771	660	583	643	721	541	341	
	CDH74	106494	3255	5391	10312	14350	15098	11803	8919	7113	8553	10667	7316	3717	
	CDH80	64547	1422	2961	6524	10105	10660	7583	4850	3375	4635	6508	4232	1692	
Wind	WSAvg	8.6	10.6	9.9	9.7	7.9	8.6	9.1	9.0	8.0	6.5	6.4	8.0	9.9	
Precipitation	PrecAvg	14.5	0.0	0.0	0.0	0.1	0.6	2.0	3.9	5.0	2.3	0.4	0.0	0.0	
	PrecMax	24.8	0.2	0.1	0.7	2.0	2.6	5.6	10.3	12.8	4.9	3.0	0.0	0.0	
	PrecMin	8.1	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.3	0.3	0.0	0.0	0.0	
	PrecStd	4.2	0.0	0.0	0.1	0.3	0.6	1.4	1.7	2.3	1.1	0.8	0.0	0.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.8	103.9	105.7	109.6	109.8	107.0	101.6	98.6	102.5	103.6	100.8	96.2	
		MCWB	60.3	62.0	63.9	67.6	68.9	71.3	73.4	75.7	71.1	66.7	64.5	61.2	
	2%	DB	93.3	99.9	104.1	107.9	107.9	104.4	98.7	95.7	99.8	101.8	99.3	93.5	
		MCWB	58.8	61.5	63.4	66.9	68.8	72.9	74.6	75.7	72.6	67.2	63.9	60.1	
	5%	DB	90.5	97.2	102.5	106.4	106.3	102.3	96.7	93.6	97.5	100.5	98.1	91.5	
		MCWB	57.3	60.4	62.9	66.4	69.8	73.5	74.6	75.7	74.0	67.6	63.1	58.7	
	10%	DB	87.8	94.3	100.6	104.8	104.4	100.2	94.8	91.7	95.3	98.9	96.1	89.3	
		MCWB	56.2	59.2	62.0	66.0	70.3	73.8	74.5	75.5	74.6	67.7	62.1	57.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	67.3	71.3	77.0	79.9	79.1	78.2	78.6	79.0	77.7	74.2	63.7
			MCDB	92.2	86.8	87.9	99.6	100.0	96.3	90.8	89.6	90.8	92.2	95.6	93.0
2%		WB	59.6	63.2	67.0	74.9	77.8	77.6	77.2	77.6	78.0	76.3	67.9	61.5	
		MCDB	91.2	96.5	98.5	97.6	97.1	94.1	89.7	88.6	89.7	90.3	91.5	91.2	
5%		WB	57.9	61.2	64.5	73.2	76.3	76.7	76.5	77.1	77.3	75.3	64.4	59.7	
		MCDB	89.0	95.4	99.3	96.2	96.4	93.3	89.0	88.1	88.9	89.0	94.4	89.5	
10%		WB	56.4	59.6	62.8	71.6	75.1	75.9	76.0	76.5	76.6	74.2	62.8	58.1	
		MCDB	86.7	93.4	98.6	95.2	95.6	92.7	88.6	87.3	88.3	88.6	94.2	87.3	
Mean Daily Temperature Range	5% DB	MDBR	25.1	25.2	24.5	23.1	20.4	18.2	16.4	15.4	17.9	21.9	25.4	25.4	
		MCDBR	27.2	27.0	26.1	24.9	22.8	20.6	18.5	17.6	20.7	23.6	26.1	27.4	
		MCWBR	12.3	11.6	10.8	10.3	9.1	6.1	4.9	4.5	6.3	8.4	10.1	11.8	
	5% WB	MCDBR	26.8	26.0	24.9	19.9	19.6	18.6	16.6	15.9	17.7	19.7	25.1	26.1	
		MCWBR	12.6	11.9	11.8	9.7	8.2	6.3	4.8	5.0	6.1	7.2	10.8	11.9	
Clear-Sky Solar Irradiance	taub	0.532	0.591	0.665	0.716	0.733	0.731	0.671	0.624	0.602	0.602	0.512	0.502		
	taud	1.802	1.637	1.504	1.474	1.472	1.513	1.669	1.753	1.776	1.704	1.808	1.841		
	Ebn at Noon	240	233	220	210	203	201	214	227	232	227	244	244		
	Edh at Noon	66	82	95	98	97	92	79	73	72	75	65	62		
All-Sky Solar Radiation	RadAvg	1742	1913	2061	2123	2088	1998	1980	1925	2013	1940	1815	1691		
	RadStd	67	79	104	81	60	84	45	62	57	53	43	62		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page